FitBuddy: AI Fitness Plan Generator using Gemini Models
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Project Documentation Phase – FitBuddy

The **Project Documentation Phase** focuses on preparing complete technical and project-related documentation for the FitBuddy – AI Fitness Plan Generator.

1. Project Overview Documentation

Document the purpose of FitBuddy, its objectives, major features, and the problem it addresses. The project uses AI to generate personalized 7-day workout plans and nutrition/recovery tips based on user information.

2. System Architecture Documentation

Document the overall architecture, including:

- **Frontend:** HTML, CSS, and Jinja2
- **Backend:** FastAPI
- **AI Layer:** Gemini Pro and Gemini Flash
- **Database:** SQLite with SQLAlchemy

3. Module Documentation

Document the major modules and their responsibilities:

- **Workout Generation Module** – Generates the 7-day workout plan.
- **Nutrition Module** – Generates nutrition/recovery tips.
- **Feedback Module** – Updates the existing plan based on feedback.
- **Database Module** – Stores user information and workout plans.
- **Routing Module** – Connects frontend, AI functions, and database.
- **Admin Module** – Displays user and plan information.

4. User Interface Documentation

Document the main web pages:

- **index.html** – User input page.
- **result.html** – Displays the personalized workout plan, nutrition tip, and feedback option.
- **all_users.html** – Admin page for viewing users and plans.

5. Database Documentation

Document the information stored in the SQLite database, including user ID, name, age, weight, fitness goal, intensity, original plan, and updated plan. The documentation should also explain that original and updated plans are preserved separately.

6. API and Route Documentation

Document the important FastAPI routes:

- `/` – Home page
- `/generate-workout` – Generates the workout and nutrition plan
- `/submit-feedback` – Updates the workout plan
- `/view-all-users` – Displays the admin view

7. Installation and Deployment Documentation

Include the required environment setup, dependencies, Gemini API configuration, and commands for running the application locally with Uvicorn. The project documentation should also mention the FastAPI `/docs` interface used for API testing.

8. Testing Documentation

Record the testing performed on user input, workout generation, nutrition tips, feedback-based updates, database storage, frontend pages, API routes, and local deployment.

9. Final Project Report

The final documentation should bring together the **project introduction, requirements, design, development, testing, deployment, screenshots, results, and conclusion**. The source project concludes that FitBuddy combines Gemini AI, FastAPI, SQLAlchemy/SQLite, and Jinja2 to provide personalized and feedback-based fitness planning.

Documentation Flow

Project Overview → Requirements → System Design → Development → Testing → Deployment → Screenshots/Results → Conclusion

This documentation provides a complete record of how **FitBuddy – AI Fitness Plan Generator using Gemini Models** was planned, designed, developed, tested, and deployed.

Continue documenting FitBuddy

- Define FitBuddy documentation deliverables
- Create FitBuddy user documentation